Project Plan

EagleEye Parking

Capstone1 - Group3

UNT-CSCE4901.502

Version 1.0

CSCE 4901- Project plan
From Project Manager

Document Control

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[1.0]	Feb 15, 2026	First final draft of Project Plan

Document Approvals

Role	Name	Signature	Date
Project Sponsor *(faculty or external sponsor)*	Jim Coffey Keitha Pearce Manos Chatzopoulos		
Project Review Group *(team members)*	Nhan Vu Thuan Nguyen Alperen Dalkir Luis Lozano Aaditya Yadav		
Project Manager	Nhan Vu Thuan Nguyen Alperen Dalkir Luis Lozano Aaditya Yadav		

1.1 Project Purpose, Goals, and Objectives

Project Purpose	
The purpose of the project is to assist UNT students, staff, faculty, and visitors by providing timely and accurate parking availability information across campus.	
Project Goals	**Project Objectives**
Goal 1: Improve Parking Efficiency on Campus	Minimize traffic congestion in high-demand parking areas
	Reduce searching time for parking
	Increase the parking reliability of parking availability information
Goal 2: Enhance User Experience within the UNT Go App	Design a clear, map-based interface
	Ensure seamless integration with the existing UNT Go Application
	Improve user adoption and satisfaction through intuitive navigation and a minimal learning curve
Goal 3: Support Data-Driven Transportation Planning	Collect and store parking utilization data for analysis and reporting
	Provide actionable insights that help UNT Transportation Services optimize parking operations
	Enable future enhancements such as predictive analytics and crowdsourced data collection.

1.2 Project Success Criteria

Category	Success Metric	Measurement Method	Target	Assigned to
Scope	Delivery of a functional prototype and logic	Comparison of final artifacts vs. the Requirements List	100% of API middleware and Dashboard config completed; UX mockups finalized	Capstone Team
Schedule	Adherence to the two-semester timeline	Milestone tracking vs Requirements Matrix (Spring and Fall 2026)	Prototype by the end of Spring; Final hand-off by Dec 2026	Capstone Team
Budget	Minimal institutional expense	Financial audit of student hosting/tools	<$500 Total	UNT Transportation Services
Quality	Data accuracy and system latency	Testing crowdsourced data against AIMS “Ground Truth.”	95%+ occupancy and accuracy; < 2 minutes data latency	Capstone Team
Stakeholder satisfaction	Approval of the final prototype	Post-presentation survey/vote by UNT Steering Group	“Approval to Proceed” from UNT CIO and Steering Group	Capstone Team
Risk management	Mitigation of technical blockers	Risk register review during the weekly Tuesday meeting	100% of “High” priority risks have a documented mitigation	Capstone Team

Resource utilization	Efficient use of student/developer hours	Weekly progress reports and contribution logs	Consistent velocity across all team members	Capstone Team Lead
Compliance	Protection of student privacy and data	Security Audit and Privacy Impact Assessment	Zero FERPA violations; 100% anonymized data	Capstone Team

1.3 Project Team

Name	Role	Responsibilities
Nhan Vu	Team Leader	Overall project coordination, stakeholder communication, and testing
Thuan Nguyen	UI/UX Designer	UI/UX wireframes and mockups for parking apps. Front-end interface design.
Alperen Dalkir	Backend Developer / API Integration	FastAPI backend development, Parkzen API integration, parking data processing pipelines, GitHub repository management
Luis Lozano	Backend Developer	FastAPI backend development, Parkzen API integration, parking data processing pipelines.
Aaditya Yadav	UI/UX Designer	UI/UX wireframes and mockups for parking apps. Front-end interface design.

1.4 Project Governance

Area	Description	Responsible Party/Role
Steering Committee	UNT stakeholders who review and approve major project decisions, including the UNT GO integration path and future rollout.	Jim Coffey (UNT Parking Services), Keitha Pearce (UNT University Systems)
Change Control Board	Evaluates and approves scope changes, requirement modifications	Jim Coffey (UNT Parking Services), Nhan Vu (Team Leader)
Compliance Auditing	This will describe how we should follow the privacy rules of the project (UNT DATA Securities rules)	Keitha Pearce (UNT University Systems)
Stakeholder Communication	Weekly standing meetings every Tuesday at 3:30 PM with all stakeholders.	Every team member and all stakeholders.
Performance Reviews	Sprint reviews at the end of each sprint cycle.	Full Project Team
Project Closure Review	Final presentation to stakeholders with demonstration of deliverables.	Full Project Team and all stakeholders.

1.5 Stakeholder Management Plan

A stakeholder management plan helps manage stakeholder expectations and address any concerns they may have. It includes the following sections.

- **Stakeholder Register:**

Name	Role	Power	Interest	Contact information
Jim Coffey	Project Sponsor (UNT Parking Services)	High	High	Jim.Coffey@unt.edu
Keitha Pearce	University Systems	High	Medium	Keitha.Pearce@unt.edu
Dr. Lotfi	Faculty Sponsor (Computer Science)	High	Medium	Lotfi.Benothmane@unt.edu
Manos Chatzopoulos	Parkzen	High	High	manos@parkzenapp.com
Nhan Vu	Team Leader	Medium	High	MonicaVu2@my.unt.edu
Thuan Nguyen	UI/UX Designer	Low	High	ThuanNguyen3@my.unt.edu
Alperen Dalkir	Backend Developer	Low	High	AlperenDalkir@my.unt.edu
Luis Lozano	Backend Developer	Low	High	LuisLozano@my.unt.edu
Aaditya Yadav	UI/UX Designer	Low	High	AadityaYadav@my.unt.edu

- **Stakeholder Map:** This visually represents the groups or individuals with a vested interest in the project and helps develop a strategy for stakeholder management.

Level of Influence		**Keep Satisfied**		**Manage Closely**	
	High		Keitha Pearce	Jim Coffey	
	Medium		Dr. Lotfi	Manos Chatzopoulos, Nhan Vu (Team Leader)	
	Low			Thuan Nguyen, Alperen Dalkir, Luis Lozano, Aaditya Yadav	
		Monitor			**Keep Informed**
		Low	**Medium**		**High**
		Level of Interest			

1.6 Scope Management Plan

The scope of the EagleEye Parking project includes the design, development, and integration of a functional parking availability system for UNT. The project will deliver a prototype that provides real-time parking visibility through a map-based interface, integrates with ParkZen data services, and aligns with UNT GO system requirements. The scope covers UI/ux design, backend API integration, database setup, system testing, and stakeholder validation. Features outside of real-time parking visibility and core system integration are considered out of scope for this phase.

Scope Baseline			
Task	**Dependencies**	**Deliverables**	**Acceptance Criteria**
Gather UI/UX requirements	Approved Project charter, stakeholder input	UI/UX requirements summary	Stakeholders confirm requirements reflect expected user needs
Create low-fidelity wireframes	Requirements finalized	wireframes for parking map & dashboards	wireframes reviewed and approved by team & sponsor
Design map - based parking view	Backing data structure defined	Interactive parking map design	Parking availability is clearly visible and understandable
Frontend integration planning	Backend API endpoints defined	UI-to-API interaction design	Fronted can successfully consume backend data
Usability review & iteration	initial UI implemented	Updated UI designs	Stakeholder feedback addressed
UI validation	Fronted integration complete	Tested UI screens	UI display accurate data without layout or usability issues

Exclusions, Assumptions and Constraints	
Exclusions	Development of physical parking sensors. Ownership or maintenance of ParkZen backend systems. Long-term production maintenance after capstone completion. Financial transaction or permit enforcement.
Assumptions	ParkZen API provides reliable and timely parking data. Stakeholders provide timely feedback during sprint reviews.
Constraints	Must align with UNT data privacy . Dependent on external system(UNT Go, ParkZen)

Milestone	Description	Delivery Date
UI Requirements Finalized	Approval of UI scope and requirements	03/15/2025
Wireframes completed	Low-fidelity UI layouts finalized	TBD
High-Fidelity Mockups	Final UI design approved	TBD
Fronted Integration Ready	UI connected to backend data	TBD
UI Testing Complete	Usability and Validation testing	TBD

1.7 Resource Requirements

For each task identified, list the resources allocated to complete the task.

Task	Resource requirements
UI/UX Requirements Gathering	Capstone team, stakeholder meetings
Wireframe & Mockup Design	Figma, UNT branding guidelines
Map interface Design	Mapping libraries, UI frameworks
Fronted UI Development	JavaScript/React
UI Testing & validation	Test users, stakeholder
Documentation	Design documentation tools

1.8 Project Schedule

Phase	Task	Dependencies	Deliverable	TImeline
Planning and Requirement Finalization	Finalized requirements, stakeholders, and scope	None	Approved requirements documents	Feb - Mar 2026
Finishing the Parkzen Dashboard	Putting in information of the lots and permits	ParkZen app and recorded lots and permits information	Completed UNT ParkZen dashboard	Feb - Mar 2026
System Design	Architecture design for EagleEye Parking (frontend, backend, database, APIs)	Approved requirements	System architecture documents, ERD, API specifications	Mar - Apr 2026
System Design	UI/UX wireframes for user and admin dashboards	Requirements and stakeholder feedback	Wireframes and UI mockups	Mar - Apr 2026
Development	Backend development: parking data ingestion API endpoints, database setup	Architecture design, ERD	Functional backend services and database schema	TBD
Integration	Integrate PArkZen-like functionality into UNT Go ecosystem	Backend + Frontend components completed	Working integrated module with real-time parking data	TBD
Integration	Implement authentication, roles, and permissions	Backened services	Unit test reports	TBD
Testing	Unit testing for backend and frontend	Completed development tasks	Unit test reports	TBD
Deployment	Deploy EagleEye Parking to staging environment	Successful system testing	Staging deployment package	TBD
Deployment	Final production deployment and documentation handoff	Staging approval	Production release + use documentation	TBD
Project Close-Out	Final presentation to stakeholders and project wrap-up	Production deployment	Final report and presentation	TBD

1.9 Project Budget

Task	Labor Costs	Material Costs	Other Costs	Budget	Actual
Requirement Gathering & Stakeholder Meetings	Internal Staff and Capstone Team time (est. 50 hrs)	None	None	$0	$0
ParkZen Dashboard Review & Data Entry	Internal Staff and Capstone Team time (est. 30 hrs)	None	None	$0	$0
ParkZen Integration (Direct Contract Scenario)	Included in vendor support	None	Annual vendor fee	$25,000 / year	TBD
Modo/UNT Go Integration (Projected Scenario)	Vendor implementation labor included	Subscription: $13,248	Implementation: $6,624	$19,872 Year 1	TBD
Backend Development (APIs, database)	Capstone Team (no cost)	Cloud hosting (if used)	None	$0 - $500 (hosting optional)	TBD
Frontend Development (Dashboard UI)	Capstone Team (no cost)	None	None	$0	$0
System Testing	Internal Staff and Capstone Team time (est. 25 hrs)	None	None	$0	$0
Deployment to Staging & Production	Internal Staff and Capstone Team time	Hosting or Modd platform	None	$0 - $500	TBD
Documentation & Training Materials	Capstone Team (no cost)	None	None	$0	$0

Important Note:

- UNT Transportation or UNIT IU currently allocates no budget for either integration path.
- Modo's estimate is not guaranteed and depends on ParkZen integration becoming available
- ParkZen's $25,000 annual cost is the only confirmed quote for a vendor
- Capstone Team labor is not billed, so labor costs are listed as $0 unless university staff time is later added
- If UNT hosts the system independently, cloud costs may apply (AWS, Azure, etc.)
- Future maintenance costs (Year 2+) depend on which integration path UNT selects

1.10 Change Management Plan

When a change request is made, use this log to track its impact, response and whether the change control board has approved it or not.

Change Control Board

Name	Role	Responsibilities

Change Order Request Form

[Download our free change request template for Excel]

Change Log

Date Identified	Request	Impact	Approval	Date Started	Date Completed

[Download a free Change Log Template for Excel]

1.11 Risk Management Plan

- **Risk Matrix:** A risk matrix is a visual representation of the potential project risks based on their likelihood and impact. It makes it easy to see what risks are pressing and what risks likely won't come to fruition.

Risk Matrix		Severity				
		Insignificant	Minor	Moderate	Major	Severe
Likelihood	Almost Certain	Medium	High	Very High	Very High	Very High
	Likely	Medium	High	High	Very High	Very High
	Possible	Low	Medium	High	High	Very High
	Unlikely	Low	Low	Medium	Medium	High

	Rare	Low	Low	Low	Low	Medium

- **Risk Log:**

Risk	Likelihood	Severity	Response	Risk Level	Risk Owner
UNT GO not accessible	Likely	Major	Independent App development	High	Keitha Pearce
No access to real sensors	Possible	Minor	Use crowd sourcing through park zen api and existing estimates	Medium	Jim Coffey
Data latency	Unlikely	Moderate	As we are working with a well known parking and gps sdk we shouldnt run into this problem as long as we implement the api in the appropriate manner	Medium	Eagle Eye Parking
Scope Creep	Rare	Moderate	We must establish rigid requirements and goals so we do not extend our scope	Low	Eagle Eye Parking
Backend Bottlenecks	Possible	Moderate	This app will in the future be rolled out to thousands of users, making sure it is easy scalable to the amount needed is key	High	Eagle Eye Parking
Inaccuracy	Possible	Major	The app revolves around occupancy, users must trust their experience or they will not return. We will run tests cases to ensure accuracy	High	Eagle Eye Parking
Privacy Concerns	Possible	Minor	As permits could be implemented into the app, we must keep user information secure and safe	Medium	Eagle Eye Parking
Team Skill	Likely	Moderate	Team is new in developing this type of app so we have to learn on the spot and make	High	Eagle Eye Parkings

			sure we implement functions correctly		
Scheduling Issues	Possible	Major	The team must be able to schedule the work in the time frame that we have for the capstone project	High	Eagle Eye Parking
Testing	Almost Certain	Severe	Team must ensure testing of all functions to prevent critical bugs that can prevent deployment	Very High	Eagle Eye Parking
Deployment	Possible	Moderate	Team will regularly deploy to a testing environment to ensure a working prototype	High	Eagle Eye Parking

Download a free risk register template for Excel

- **Risk Mitigation Strategies:** Various techniques can be used to reduce the impact or likelihood of a risk occurring. Some examples include risk reduction, risk avoidance or risk acceptance.
- **Assumptions and Constraints:** List any assumptions you have about the project. Then note the project constraints. Finally, in an appendix, you'll want to attach the supporting documentation, such as the project schedule, business case, feasibility study, project charter, etc.

1.12 Quality Management Plan

We also include a quality management plan in our project plan template. It's made up of the following parts.

- **Quality Standards:** These are the criteria and benchmarks that measure the quality of a project's processes and deliverables. They enable the project to meet stakeholder expectations.
- **Quality Assurance Guidelines:** Next are quality assurance guidelines that help define quality standards and establish quality control procedures. These guidelines can help improve customer and stakeholder satisfaction.
- **Quality Control Procedures:** List the specific actions that you'll take to ensure the product, service or process meets the predetermined quality standards.

Task or Deliverable	Quality Standards	Quality Assurance Guidelines	Quality Control Procedures
Requirements Documentation(SRS)	Requirements are clearly established, complete, have test cases and are quantifiable	Requirements align with project scope and accepted project charter to align with stakeholder vision	Completion of test cases and requirement checklist to ensure deliverable is what was promised as a requirement
Architecture	Project Architecture supports future scalability and is reliable at said size	Use industry standards in map derived apps to help team deliver a scalable app	Ensure architecture holds up through all phases of development. Verify front end and backend both follow standards

Database systems	Maintain data integrity, schema supports scalability and accuracy	Use schema with known success rates, assure data flow is efficient and up to standard	Minimize redundancy, optimize performance make sure it is a clear and well documented schema
API integration	API responses are accurate and secure and the performance is up to industry standards	Follow REST standards for API's, introduce error handling	Test API using test cases,verify correct responses and status codes
Map View	Parking availability is clearly labeled and shown and easy to interpret	Use UNT provided UI theme guidelines to fit app standard and use colors and labels which users can easily understand	Visual inspections with possible early user testing to gauge feedback on user experience
Frontend View	Information is clear and UI follows a clear theme with readable information	Maintain consistent design language across app	Functional testing, verify frontend displays correct backend data
Data Freshness	Users can see how recently parking lot occupancy was updated to gauge their decisions	Request timestamps from API so we can convey that data freshness to users	Test cases with old data and fresh data, warnings about data delay
Security	API's are secure and not open to malicious actors	Setup admin controls, follow secure coding practices	Security testing, verify access and authentication
Performance	App is seamless and load times are acceptable	Improve performance of systems as phases are completed, ensure new functions do not slow down existing systems	Test with comparable loads to peak usage time
Documentation	App is thoroughly documented so team can hand over app to UNT if needed	Update documentation as we complete all function, remove stale information	Testing by foreign parties with documentation to ensure correct function documentation

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1.13 Appendix

This section is subjected to change as needed.